

Impact of Dose Rate on Clinical Course in Uveal Melanoma after Brachytherapy with Ruthenium-106

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Background and Purpose: It has been suggested that the actual dose rate of an irradiating source may be a distinct influencing factor for the biological effect after brachytherapy with ruthenium-106 for uveal melanoma. The purpose of this study was to investigate a hypothesized impact of the dose rate on the clinical and echographic course after brachytherapy.

Patients and Methods: In total, 45 patients were included in this retrospective study. According to the actual dose rate, two groups were defined: group 1 with a dose rate < 4 Gy/h and group 2 with a dose rate ≥ 4 Gy/h. Regarding age, tumor height, basal diameter, scleral and apical dose, differences between the groups were not significant. Clinical parameters, including early and late side effects, and echographic courses were compared.

Results: A significantly lower metastatic rate was found in group 2. Using univariate Cox proportional hazards regression, only dose rate predicted metastatic spread significantly ($p < 0.05$), while in a multivariate analysis, using age at the time of treatment, greatest tumor height and greatest basal diameter as covariates, the variable dose rate was of borderline significance ($p = 0.077$). Patients in group 2 had more early side effects and more pronounced visual decline, but these differences were of borderline significance with p -values of 0.072 and 0.064, respectively.

Conclusion: These data suggest that a higher dose rate may confer a lower risk for metastatic spread, but may be associated with more side effects and more pronounced visual decline.

Key Words: Uveal melanoma · Ruthenium-106 · Dose rate

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Einfluss der Dosisrate auf den klinischen Verlauf beim uvealen Melanom nach Brachytherapie mit Ruthenium-106

Hintergrund und Ziel: Es wurde vermutet, dass die Dosisrate einer Strahlenquelle einen maßgeblichen Faktor für den biologischen Effekt nach Brachytherapie mit Ruthenium-106 beim uvealen Melanom darstellt. Ziel dieser Studie war die Untersuchung des Einflusses der Dosisrate auf den klinischen und echographischen Verlauf nach Brachytherapie.

Patienten und Methodik: Insgesamt wurden 45 Patienten in diese retrospektive Studie eingeschlossen. Entsprechend der Dosisrate zum Zeitpunkt der Brachytherapie wurden zwei Gruppen definiert: Gruppe 1 mit einer Dosisrate < 4 Gy/h und Gruppe 2 mit einer Dosisrate ≥ 4 Gy/h. Bezüglich des Alters, der Tumorphöhe, des basalen Durchmessers sowie der Sklerakontakt- und apikalen Dosis bestand kein signifikanter Unterschied zwischen den beiden Gruppen. Klinische Parameter inklusive früher und später Nebenwirkungen sowie der echographische Verlauf wurden verglichen.

Ergebnisse: Es fand sich eine signifikant niedrigere Metastasierungsrate in Gruppe 2. Mit der univariaten Cox-Regression zeigte sich nur für die Dosisrate ein signifikanter Zusammenhang ($p < 0,05$), während der Zusammenhang in der multivariaten Analyse mit Alter, größter Tumorphöhe und größtem Tumordurchmesser als Kovariablen grenzwertig war ($p = 0,077$). Patienten der Gruppe 2 hatten eine höhere Rate an frühen Nebenwirkungen und einen größeren Visusverlust, wobei die Unterschiede grenzwertig signifikant waren ($p = 0,072$ bzw. $p = 0,064$).

Schlussfolgerung: Diese Daten lassen vermuten, dass eine höhere Dosisrate ein niedrigeres Metastasierungsrisiko aufweist, aber mit einer höheren Rate an Nebenwirkungen und größerem Visusverlust assoziiert ist.

Schlüsselwörter: Aderhautmelanom · Ruthenium-106 · Dosisrate

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Introduction

Since the introduction of plaque brachytherapy with cobalt-60 (^{60}Co) by Moore and his successor Stallard, different low-energy isotopes were established as a choice of therapy for small- to medium-sized uveal malignant melanoma [20, 24]. In 1964, the isotope ruthenium-106 (^{106}Ru) was first applied by Lommatzsch & Vollmar because of its favorable dose gradient and is nowadays in common use throughout Europe and other parts of the world [16]. Only recently, the prospective Collaborative Ocular Melanoma Study (COMS) confirmed brachytherapy as a safe alternative to enucleation, but iodine-125 (^{125}I) was used in this study [5]. For tumors located in close proximity to the optic nerve head or the fovea, alternative therapies like gamma knife stereotactic radiotherapy, proton therapy and linear accelerator-based stereotactic radiotherapy have been used [3, 4, 9, 14, 18].

The radiation energy absorbed per time unit is defined by the dose rate (i.e., Gy/h) and is an important factor for the radiobiological effect of the isotope. The dose rate, however, changes because of the relatively short half-life of ^{106}Ru , which is 373.59 days. It has been suggested from both experimental and retrospective studies that higher dose rates may lead to increased side effects and increased local tumor control. Using ^{60}Co plaques, Rotman et al. first reported higher local treatment failure after low-dose-rate brachytherapy [22], while Beitler et al., using ^{60}Co plaques as well, found no correlation between dose rate and local treatment failure [2]. In 1986, Lommatzsch et al. observed more exudative and hemorrhagic side effects after a short period of irradiation with ^{106}Ru , but found no difference in terms of tumor regression [17]. Klein-eidam & Guthoff, using ^{106}Ru , reported an increased metastatic rate after high-rate brachytherapy in tumors measuring < 10 mm in diameter [12].

To investigate a hypothesized impact of the dose rate, we retrospectively compared the clinical and echographic course in posterior uveal melanoma after brachytherapy with ^{106}Ru .

Patients and Methods

45 patients, who received brachytherapy with ^{106}Ru for uveal melanoma in our department between 1990 and 2003, were included in this retrospective study. Patients were allocated according to the actual dose rate of the ^{106}Ru plaque to group 1 (dose rate < 4 Gy/h) or group 2 (dose rate $\geq$ 4 Gy/h). Group 1 comprised 25 patients (15 female patients), while group 2 comprised 20 patients (eleven female patients). Criteria for diagnosing uveal melanoma were clinical appearance typical of uveal malignant melanoma, tumor thickness > 3 mm, serous retinal detachment, and echographically documented tumor growth. According to the site, tumors were classified as either juxtapapillary, macular, ciliary, or peripheral melanoma. Preoperatively, all patients underwent slit-lamp biomicroscopy, testing for best-corrected visual acuity using the Snellen chart, and echographic examination using both A and B modes of standardized ultrasound (B-Scan-S, Biovision, Clermont-Fer-

rand, France, or Cinescan S, Quantel Medical, Clermont-Fer-rand, France) [23]. Particular dose rates were calculated by means of an in-house developed software tool according to the source certificate provided by the manufacturer.

Postoperatively, echographic measurements were obtained every month for the first 3 months, every 3 months thereafter for the 1st year, and then every 6 months. Local tumor control was achieved, if tumor height decreased continuously without applying a second therapy and without echographic evidence of regrowth. Time and method of secondary treatment (enucleation, brachytherapy with ^{106}Ru , transpupillary thermotherapy [TTT], local resection, gamma knife stereotactic radiotherapy) were recorded. Metastatic screening was performed annually by ultrasound or computed tomography examination of the liver, liver function tests, and X-ray of the lungs. Best-corrected visual acuity was tested at least every 12 months postoperatively. Immediate postoperative increase of concomitant serous retinal detachment or development of choroidal detachment were classified as early side effects. Radiation retinopathy was diagnosed as soon as retinal microaneurysms, hemorrhages, exudation, cotton-wool spots, or retinal or optic disk neovascularization were found. Duration of follow-up until the last examination at our department was recorded in months.

Statistics

Descriptive statistics were used to calculate frequencies and percentage of discrete variables. Proportions of groups were compared by χ^2 -test. Means were compared using Mann-Whitney test. Cox proportional hazards model was used to calculate prognostic values. The criterion for statistical significance was $p < 0.05$. Statistical analysis was done using SPSS 10.0 for windows (SPSS, version 10.0, Chicago, IL, USA).

Results

Mean age was 65.2 years (range 49.9–80.2 years) in group 1 and 58.9 years (range 35.5–80.9 years) in group 2. Preoperative tumor characteristics are shown in Table 1. Echographically, no extraocular extension was observed. Means of the

Table 1. Preoperative tumor characteristics in patients of group 1 and group 2.

Tabelle 1. Präoperative Tumorparameter der Patienten in Gruppe 1 und Gruppe 2.

	Group 1 (n = 25)	Group 2 (n = 20)
Tumor height (mm) [mean (range)]	5.2 (3.1–9.9)	5.5 (3.0–8.4)
Largest diameter (mm) [mean (range)]	10.3 (5.2–16.4)	11.6 (6.3–17.0)
Juxtapapillary [n (%)]	2 (8)	4 (20)
Macular [n (%)]	2 (8)	1 (5)
Peripheral [n (%)]	17 (68)	13 (65)
Ciliary [n (%)]	4 (16)	2 (10)

Table 2. Radiation parameters, preoperative best-corrected visual acuity, and duration of follow-up in patients of group 1 and group 2.

Tabelle 2. Bestrahlungsparameter, präoperativer bestkorrigierter Visus und Dauer der Nachbeobachtung der Patienten in Gruppe 1 und Gruppe 2.

	Group 1 (n = 25) Mean (range)	Group 2 (n = 20) Mean (range)
Dose rate (Gy/h)	2.6 (1.2–3.9)	5.9 (4.0–10.1)*
Apical dose (Gy)	120 (96–210)	106 (100–150)
Scleral dose (Gy)	841 (500–1,100)	802 (370–1,250)
Best-corrected visual acuity (Snellen equivalent)	0.5 (0.05–1.0)	0.5 (hand motion – 1.0)
Duration of follow-up (months)	60.4 (16–110)	63.2 (19–114)

*p < 0.001

calculated dose rate at the time of application, apical dose, scleral dose, best-corrected visual acuity preoperatively, and duration of follow-up are shown in Table 2. With the exception of mean dose rate no statistically significant difference was observed between both groups.

Local tumor control was achieved in 22 patients (88%) in group 1 and 18 patients (80%) in group 2, respectively. The difference was statistically not significant. In group 1, two patients received TTT and one patient underwent gamma knife stereotactic radiotherapy as second therapy, while in group 2, one patient received gamma knife stereotactic radiotherapy and the other was treated with local resection of the tumor.

Local recurrence was observed in six patients (24%) in group 1 after a mean follow-up duration of 42.7 months (range

19–80 months) and four patients (20%) in group 2 after a mean follow-up duration of 27.3 months (range 15–39 months), respectively. The difference was statistically not significant. In group 1, three patients were treated with enucleation, two received a second brachytherapy with ^{106}Ru , and one patient was treated with TTT. In group 2, two patients underwent TTT, one patient was treated with enucleation, and one received a second brachytherapy with ^{106}Ru .

Using the Kaplan-Meier actuarial method, the overall actuarial rate of metastatic spread was 13.3% at 5 years. In five patients (20%) of group 1, metastatic spread to the liver was detected, while no metastatic spread was found in patients of group 2 ($p < 0.034$). In two patients, metastatic spread was detected after local recurrence had occurred. Using univariate Cox proportional hazards regression, only dose rate predicted metastatic spread significantly ($p < 0.05$), while in a multivariate analysis, using age at the time of treatment, greatest tumor height and greatest basal diameter as covariates, the variable dose rate was of borderline significance ($p = 0.077$).

Echographic courses expressed in means of percentage of the initial tumor height of the first 3 months are shown in Figures 1 and 2. Differences were statistically not significant. Echographic course of the first 2 years showed an approximately similar regression in both groups (data not shown). Relative tumor reduction after 12 and 24 months was 35% and 45% in group 1, and 30% and 43% in group 2, respectively.

Incidence of early side effects, radiation retinopathy, and mean visual decline expressed in loss of lines is shown in Table 3. Differences in the incidence of early side effects and

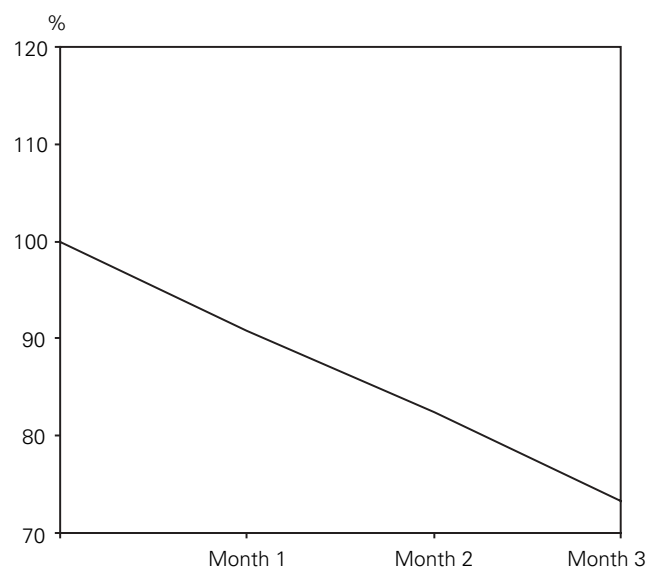


Figure 1. Echographic course of the first 3 months in group 1 expressed in means of percentage of the initial tumor height.

Abbildung 1. Echographischer Verlauf der ersten 3 Monate in Gruppe 1, dargestellt als prozentualer Durchschnitt bezogen auf die initiale Tumorphöhe.

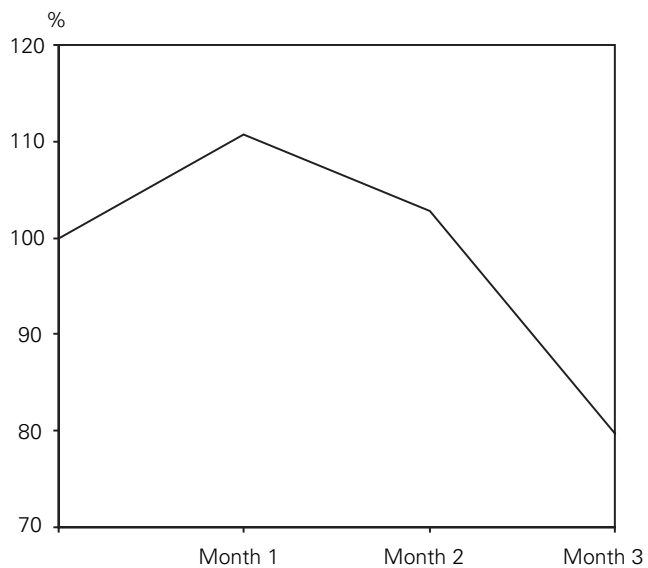


Figure 2. Echographic course of the first 3 months in group 2 expressed in means of percentage of the initial tumor height.

Abbildung 2. Echographischer Verlauf der ersten 3 Monate in Gruppe 2, dargestellt als prozentualer Durchschnitt bezogen auf die initiale Tumorphöhe.

Table 3. Incidence of early side effects, radiation retinopathy and visual decline expressed in loss of lines in patients of group 1 and group 2. SD: standard deviation.**Tabelle 3.** Inzidenz der frühen Nebenwirkungen, radiogenen Retinopathie und Visusverschlechterung in Zeilenverlust der Patienten in Gruppe 1 und Gruppe 2. SD: Standardabweichung.

	Group 1 (n = 25)	Group 2 (n = 20)
Early side effects [n (%)]	5 (20)	9 (45)
Radiation retinopathy [n (%)]	4 (16)	5 (25)
Visual decline after one year (mean $\pm$ SD)	1.7 $\pm$ 3.9	4.6 $\pm$ 6.1
Visual decline after 2 years (mean $\pm$ SD)	2.7 $\pm$ 4.9	5.1 $\pm$ 6.2

mean visual decline after 1 year were of borderline significance ($p = 0.072$ and $p = 0.064$, respectively). Serous retinal and choroidal detachments resolved spontaneously in all patients.

Discussion

In the present study, we found a significantly lower metastatic rate in the group with a dose rate ≥ 4 Gy/h. Furthermore, the local recurrence rate was lower in the group with a dose rate ≥ 4 Gy/h without reaching statistical significance. Hall demonstrated that sublethal damage repair occurs with lower dose rates allowing cell recovery during irradiation, which might explain the observed difference in our study [8]. Furthermore, in an in vitro study using human melanoma cells (MeWo), Sterzing et al. reported increasing survival rates after irradiation with lowered dose rates [25]. Only few studies investigated the role of the dose rate in brachytherapy for uveal melanoma. In their multivariate analysis of the impact of time and dose factors on local failure, Quivey et al. reported a significant correlation between local failure and lower radiation dose rate after brachytherapy with ^{125}I [21]. In an animal model with experimental Greene choroidal melanoma, Yang et al. provided evidence that ^{125}I plaques with higher dose rates showed increased damage to normal chorioretinal tissues, but were more effective in terms of tumor eradication [27]. As regards radiotherapy with ^{106}Ru , Lommatzsch et al. found significantly more damage to normal chorioretinal tissue after high-dose irradiation of rabbit eyes, while they observed no difference in terms of tumor control in a retrospective study [17].

The results of the present study are in contrast to the findings of Kleineidam & Guthoff, who reported an increased metastatic rate after high-rate brachytherapy with ^{106}Ru solely in tumors measuring < 10 mm in diameter [12]. Furthermore, Kleineidam et al. performed an experimental study with the rhabdomyosarcoma of the R1H rat, which was implanted subcutaneously into the back of syngeneic WAG/Rij albino rats and then treated with ^{106}Ru brachytherapy using three different dose rates and identical total doses. Interestingly, a dose rate of 1.3 Gy/h led to a complete local tumor control, while no

local control was observed after brachytherapy with a dose rate of 6 Gy/h and 0.8 Gy/h [13]. This was assumed to be caused by incomplete reoxygenation in the case of brachytherapy with a dose rate of 6 Gy/h, and by proliferation of tumor cells under treatment in the case of brachytherapy with a dose rate of 0.8 Gy/h. The volume doubling time of the rhabdomyosarcoma of the R1H rat, however, is rather fast (3 days at a volume of 1 cm^3) compared to the volume doubling time of primary uveal melanoma (23–540 days depending on the histological type), which makes the study by Kleineidam et al. difficult to interpret [1, 26].

The cell type of the melanoma and certain vascular patterns are suggested to be important prognostic factors [6, 19]. As these features were not assessed in the present study, they represent a potential source of bias. We recognize that the greatest limitation of our study is the limited sample size.

We report a tendency of increased rate of early and late side effects and greater visual decline after 1 and 2 years in the high-dose-rate group. As higher dose rates may lead to more tumor cell kill, more damage to the normal tissues of the eye can also be expected. As mentioned before, Yang et al. and Lommatzsch et al. reported increased damage to normal ocular tissues after high-dose-rate radiation [17, 27]. Moreover, Gündüz et al. reported a significant correlation between radiation retinopathy and higher dose rate, while Jones et al. reported a significant correlation between visual decline and higher dose rates to the macula [7, 10]. Interestingly, in a retrospective study of gamma knife stereotactic radiotherapy comparing high- versus low-dose radiosurgery, the authors reported an increased rate of side effects in the high-dose group [15].

In addition, we describe different echographic courses after high- or low-dose brachytherapy in the first 3 months. Tumors treated with high-dose-rate brachytherapy exhibited an initial increase of tumor height, which may be due to increased vascular leakage. However, we found no significant difference in terms of relative tumor height after 2 years (55% and 57%, respectively). Kaiserman et al. reported a tumor height of approximately 61% of the tumor height at brachytherapy after 2 years, which is comparable to our results [11].

Conclusion

Our results suggest that brachytherapy with high dose rate may confer a lower risk for metastatic spread, but may be associated with more side effects and more pronounced visual decline. It seems reasonable to further investigate the impact of the dose rate in the brachytherapy of uveal melanoma in a prospective trial.

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